

SOHO (Small Office/Home Office)

Network Design using Cisco Packet Tracer

Project Background

XYZ Company is a growing company planning to open a new branch in Bonalbo, Eastern Australia. The branch will operate independently from the company's headquarters and requires its own network infrastructure.

As the assigned network engineer, the objective of this project is to design and implement a small office network using Cisco devices based on the given requirements.

Project Requirements

- Use one Cisco router and one Cisco switch.
- Create three departments:
 - Admin/IT
 - Finance/HR
 - Customer Service/Reception
- Configure a separate VLAN for each department.
- Provide wireless access for each department.
- Configure automatic IPv4 address assignment (DHCP).
- Allow communication between all departments.
- Use the assigned network 192.168.1.0/24.

Purpose

The purpose of this project is to practice designing and configuring a small office network using Cisco Packet Tracer while applying networking concepts such as VLANs, DHCP, wireless networking, and inter-device communication.

Devices Used

- Cisco Router
- Cisco Catalyst 2960 Switch
- Wireless Access Points
- Desktop PCs
- Wireless PCs
- Tablet
- Smartphone